Comprehensive Full-Stack Guide: Architecture, Logging, CI/CD & Java Roadmap

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> **Executive Overview:** This guide explains how client-server deployment works, how to get a permanent static URL, where to monitor API logs, how CI/CD pipelines auto-recover on failures, and provides a full roadmap for enterprise Java full-stack projects.

1. Root Cause: Why Did the Disconnect Happen?

```
+-------------------------------------------------------------------------+ | CLIENT (Frontend Browser)
| | - HTML, CSS, JavaScript (React / Vite bundle) | | - Runs strictly inside the visitor's web browser
| | - Does NOT have direct database access or file system access | +-----------------------------------
-----------------------------------+ | HTTP / REST Requests (fetch / axios) v +--------------------
-------------------------------------------------+ | SERVER (Backend Host) | | - Node.js (Express)
or Java (Spring Boot) | | - Manages database transactions, authentication, and business logic | | -
Listens on a dedicated network port (e.g. 5000 or 8080) | +----------------------------------------
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```

- **Port 3000 was static-only:** `npx serve` served HTML/CSS. When React requested `/api/students`, it returned `index.html` (causing JSON syntax errors).
- **Cloudflare pointed only to Port 3000:** Port 5000 (Backend API & Database) was unreachable.
- **Unified Port 5000 Solution:** Express on Port 5000 now serves the React build from `/` AND all REST APIs from `/api/*` on the same port!

2. How to Get a Constant / Permanent URL (No Random URLs)

When using `cloudflared tunnel --url http://localhost:5000`, Cloudflare generates a random temporary URL (Quick Tunnel). Here is how to get a **constant, unchanging URL**:

> **Method A: LocalTunnel with Fixed Subdomain (Instant & Zero Configuration)**
>
> Run the following command in PowerShell. It gives you the exact same URL every single time:
>
> `npx localtunnel --port 5000 --subdomain aegisedu-portal`
> **Permanent URL:** `https://aegisedu-portal.loca.lt`

> **Method B: Cloudflare Named Tunnel (Official Enterprise Method)**
>
> 1. Login: `cloudflared tunnel login`
> 2. Create tunnel: `cloudflared tunnel create aegis-prod`
> 3. Route to your domain or subdomain: `cloudflared tunnel route dns aegis-prod portal.yourdomain.com`
> 4. Run: `cloudflared tunnel run aegis-prod`

> **Method C: Permanent Free Cloud Deployment (Render / Railway)**
>
> Deploy directly to Render or Railway. You get a permanent, 24/7 online URL (e.g., `https://aegisedu.onrender.com`) that never changes and doesn't require your laptop to be awake.

3. Where and How to View API Request Logs

You can monitor API activity in three different places:

- **1. Live Terminal (Real-Time Console):** In the terminal running `npm start`, every incoming request is logged live:
  `⚡ [18:55:12] GET /api/stats -> HTTP 200 (12ms)`
  `⚡ [18:55:15] GET /api/students?page=1 -> HTTP 200 (8ms)`
  `⚠️ [18:55:30] POST /api/auth/login -> HTTP 401 (4ms)`
- **2. In the Web Database Manager:** Open `http://localhost:5000/` → Click the **Database** tab → Open the **audit_logs** table to view system events and diagnostic records.
- **3. Browser Developer Tools:** Press `F12` → Click the **Network** tab → Filter by `Fetch/XHR` to see live requests, response times, headers, and JSON payloads.

4. CI/CD Pipeline & Automated Recovery

A Continuous Integration & Continuous Deployment (CI/CD) pipeline ensures your app automatically tests, builds, and redeploys without breaking the live URL:

```
Developer Pushes Code to GitHub ('main' branch) | v +----------------------------------------------
----------------------+ | GITHUB ACTIONS CI/CD PIPELINE | | 1. Checks out repository | | 2. Installs
dependencies (npm install) | | 3. Runs Frontend Build (npm run build) | | 4. Runs Automated Tests &
Code Integrity Checks | +-------------------------------------------------------------------------------+ | |
| IF BUILD FAILS | IF BUILD PASSES v v +-------------------------------------+ +--------------------
-------+ | Pipeline Halts & Alerts Developer | | Automatically Redeploys to | | Live URL Stays on Last
Working Version| | the SAME Constant URL | | ZERO DOWNTIME FOR USERS | | (Render / Railway / Server)|
+-------------------------------------+ +---------------------------+
```

Auto-Recovery with Process Supervisors (PM2)

If your server crashes due to an unhandled exception or memory issue, a process supervisor restarts it in milliseconds so your tunnel stays alive:

`npm install -g pm2`

`pm2 start server/server.js --name "aegisedu" --watch`

`pm2 logs aegisedu` (View live logs)

5. Enterprise Java Full-Stack Project Roadmap

Phase	Topics & Deliverables	Tools & Technologies
1. Core Backend	REST API, 3-tier architecture (Controller, Service, Repository), DTOs	Java 21, Spring Boot 3, Spring Web
2. Database Layer	JPA Entities, Hibernate ORM, Automated CRUD, Zero-Admin H2 Database	Spring Data JPA, H2 / MySQL
3. Security	Stateless authentication, JWT tokens, Role-based permissions, CORS	Spring Security 6, jjwt
4. Unified Packaging	Build React (`npm run build`) → Copy to `src/main/resources/static/` → Package into single JAR	Maven, frontend-maven-plugin
5. Deployment	Run `java -jar app.jar` on port 8080 → Expose with Cloudflare Tunnel or Docker	Cloudflare Tunnel, Docker

6. Node.js vs. Java Full-Stack Cheat Sheet

Feature	Node.js (Your Current Project)	Java Full-Stack (Spring Boot)
Backend Engine	Express.js (Port 5000)	Spring Boot 3 (Port 8080)
Zero-Install DB	Native SQLite (`server/student.db`)	Embedded H2 (`./data/studentdb.mv.db`)
Unified Static Path	`app.use(express.static('../react/dist'))`	`src/main/resources/static/`
Packaging	`npm start`	`mvn package` → `java -jar app.jar`
Tunnel Command	`cloudflared tunnel --url http://localhost:5000`	`cloudflared tunnel --url http://localhost:8080`

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